

CS Rubric – Predicting Recommendation for Airports Based on Airport Reviews and determining key factors for Recommendation

DS 4002 – Spring 2026
Individual Assignment

Submission format:

- Submit to Canvas with a link to your GitHub Repository

Why am I doing this? This case allows you to explore the Logistic Regression Classifier model and the Linear Regression models through the lense of a real-life case. By working through this case, you will be able to see the value of these models for everyday inputs, like surveys. You will also gain the experience of taking raw outputs and converting them in digestible information for related stakeholders.

- Course Learning Objective: prepare findings for presentation to your peers.

What am I going to do? You will first read through the dataset and to understand its contents. Then you will use the data set to create a Logistic Regression – Classifier model for the text to find the top 10 most common words for a customer that would recommend an airport and the top 10 words for customers that would not recommend an airport. You will then create a Linear Regression model to determine feature importance for the difference aspect within an airport. Lastly, you will extract from your results a conclusion and a recommendation for an airport of your choice as to how to improve their rating using the models you've used.

Tips for success:

- Talk to the instructors. This is a creative assignment, and you are allowed to show ideas to people for comment.
- Talk to your fellow students. This is a creative assignment and you are allowed to show ideas to people for comment.

How will I know I have Succeeded? You will meet expectations on CS - Predicting Recommendation for Airports Based on Airport Reviews and determining key factors for Recommendation when you follow the criteria in the rubric below.

Formatting	• The GitHub Repository should contain the following documents: - README.md file - LICENSE.md file - Data folder - Coding Script for Linear Regression Model - Coding Script for Logistic Regression- Classifier Model - AI Acknowledgment if used - Output File

	- Conclusion and Recommendation File
README.md	• <u>Goal:</u> Provide context into what your project is • Should include the following - Title, Date, Author - Summary of Project in. one paragraph - Data Set Title - Variables for each model
LICENSE.md	• <u>Goal:</u> Explain intent of the citation of the repository. • THE MIT license is usually the appropriate one
Data Folder	• <u>Goal:</u> Provide easily accessible file to view data • Data set used
Coding Scripts	• <u>Goal:</u> View how code was written so it can be reviewed and replicated • Linear Regression Model • Logistic Regression – Classifier Model
AI Acknowledgment	• <u>Goal:</u> Provide insight if and how AI was used to aid your work
Output File	• <u>Goal:</u> Singular source to view results • <u>Should include the following:</u> - Visuals from models - Statistical summaries from models
Conclusion and Recommendation	• <u>Goal:</u> One easily accessible location for your conclusion and next steps • <u>2 paragraphs</u> - What the data has resulted and what conclusions you have drawn from it for the first paragraph - How you would recommend a specific airport of your choosing improve their airport operations with specific actionable steps

Acknowledgements: Special thanks to Jess Taggart from UVA CTE for coaching on making this rubric. This structure is pulled from [Streifer & Palmer \(2020\)](#).